Evaluate yourself

2. Atomic number : 120

IUPAC temporary symbol : Unbinilium

IUPAC temporary symbol : Ubn

Possible electronic configuration :[Og] $8s^2$

3. Electronic Configuration : $(n-1)d^2 ns^2$

for n = 5, the electronic configuration is,

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 4d^2\ 5s^2$ Atomic number : 40

4th group 5th period (d block element) = Zirconium

4. Electronic Configuration of Aluminium

$_{13}Al\ \underbrace{1s^2}_{(n-2)}\ \underbrace{2s^2\ 2p^6}_{(n-1)}\ \underbrace{3s^2\ 3p^1}_{n}$

Group	no. of electrons	Contribution of each electron to 'S' value	Contribution of a particular group
n	2	0.35	0.70
(n – 1)	8	0.85	6.80
(n – 2)	2	1	2.00
			9.50

∴ Effective nuclear charge = Z – S

= 13 – 9.5

$(Z_{eff})_{Al}$ = 3.5

Electronic Configuration of chlorine

$\underbrace{1s^2}_{n-2}\ \underbrace{2s^2\ 2p^6}_{n-1}\ \underbrace{3s^2\ 3p^5}_{n}$

Group	no. of electrons	Contribution of each electron to 'S' value	Contribution of a particular group
n	6	0.35	2.1
(n – 1)	8	0.85	6.8
(n –2)	2	1	2
		S =	10.9

∴ Effective nuclear charge = Z – S

= 17 – 10.9

$(Z_{eff})_{cl}$ = 6.1

$(Z_{eff})_{cl} > (Z_{eff})_{Al}$ and. Hence $r_{cl} < r_{Al}$.

5.

X^{3+}, Y^{2+}, Z^- are isoelectronic.

∴ Effective nuclear charge is in the order $(Z_{eff})_{Z^-} < (Z_{eff})_{Y^{2+}} < (Z_{eff})_{X^{3+}}$ and

Hence, ionic radii should be in the order

$r_{Z^-} > r_{Y^{2+}} > r_{X^{3+}}$

∴ The correct values are,

Species	Ionic radii
Z^-	136
Y^{2+}	64
X^{3+}	49

6. **Noble gases :** Ioniation energy ranging from 2372 $KJmol^{-1}$ to 1037 kJ mol^{-1}.
For element X, the IE_1, value is in the

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